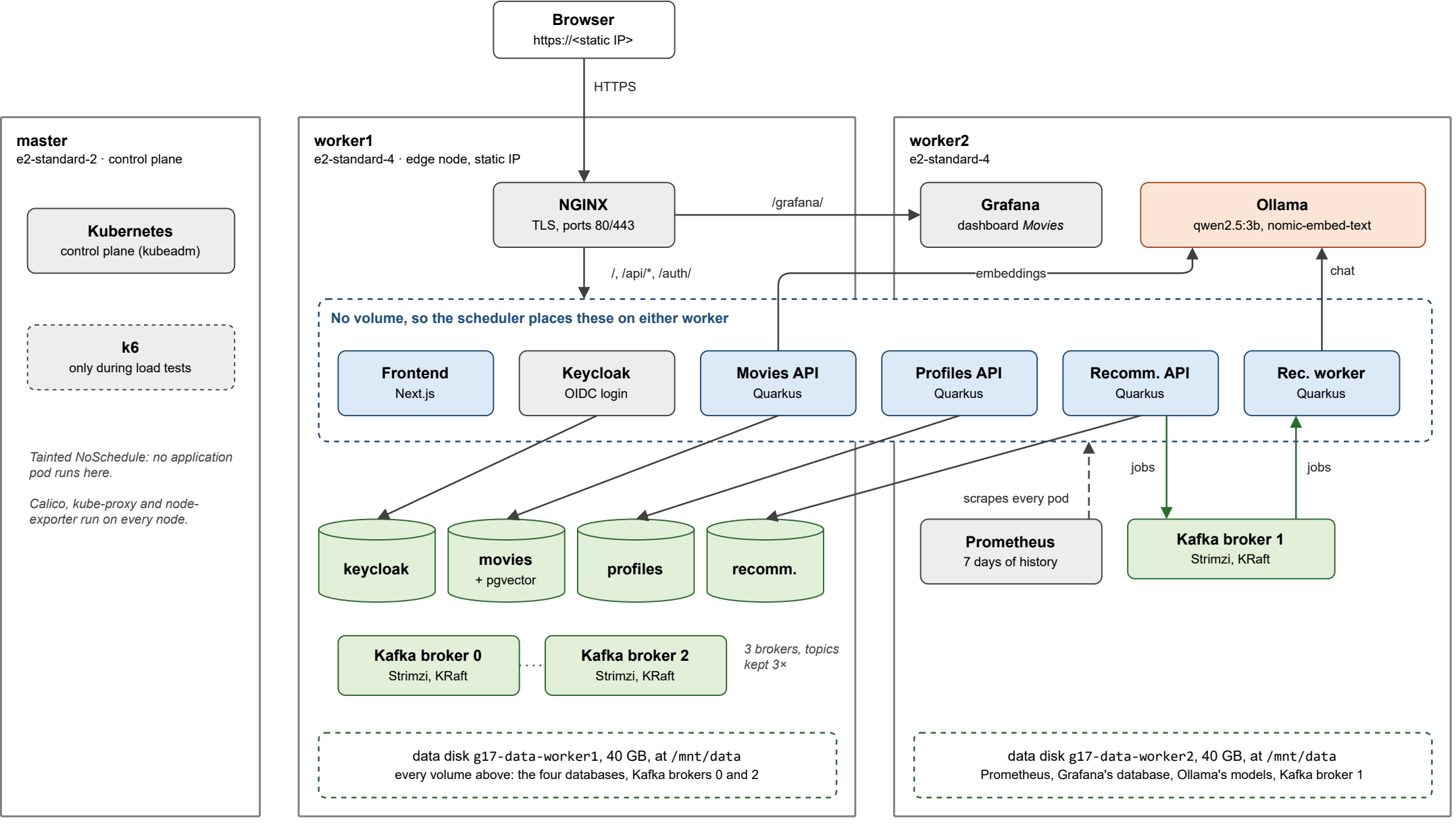

Browser
https://<static IP>
HTTPS
master
e2-standard-2 · control plane
Kubernetes
control plane (kubeadm)
k6
only during load tests
Tainted NoSchedule: no application pod runs here.
Calico, kube-proxy and node-exporter run on every node.
worker1
e2-standard-4 · edge node, static IP
NGINX
TLS, ports 80/443
/grafana/
/, /api/*, /auth/
worker2
e2-standard-4
Grafana
dashboard Movies
Ollama
qwen2.5:3b, nomic-embed-text
embeddings
chat
No volume, so the scheduler places these on either worker
Frontend
Next.js
Keycloak
OIDC login
Movies API
Quarkus
Profiles API
Quarkus
Recomm. API
Quarkus
Rec. worker
Quarkus
jobs
jobs
scrapes every pod
keycloak
movies
+ pgvector
profiles
recomm.
Prometheus
7 days of history
Kafka broker 1
Strimzi, KRaft
Kafka broker 0
Strimzi, KRaft
Kafka broker 2
Strimzi, KRaft
3 brokers, topics kept 3×
data disk g17-data-worker1, 40 GB, at /mnt/data
every volume above: the four databases, Kafka brokers 0 and 2
data disk g17-data-worker2, 40 GB, at /mnt/data
Prometheus, Grafana's database, Ollama's models, Kafka broker 1
application
database / Kafka
inference engine
platform, operations
request
Kafka message
monitoring